

OOPs assignment 8

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Batch:2o13

Function Template

1. Write a function template to swap two variables of any data type.

```
ain.cpp
1  #include <iostream>
2  using namespace std;
3
4  template <typename T>
5  void swapValues(T &a, T &b) {
6      T temp = a;
7      a = b;
8      b = temp;
9  }
10
11 int main() {
12     int x = 5, y = 10;
13     swapValues(x, y);
14     cout << x << " " << y;
15 }
```

```
10 5

...Program finished with exit code 0
Press ENTER to exit console.
```

2. Write a function template to find the minimum element in an array.

```
arr.cpp
1  #include <iostream>
2  using namespace std;
3
4  template <typename T>
5  T findMin(T arr[], int n) {
6      T min = arr[0];
7      for(int i=1;i<n;i++)
8          if(arr[i] < min)
9              min = arr[i];
10     return min;
11 }
12
13 int main() {
14     int arr[] = {4,2,7,1};
15     cout << findMin(arr,4);
16 }
```

```
input
1
...Program finished with exit code 0
Press ENTER to exit console.
```

3. Write a function template to sort an array using bubble sort.

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  template <typename T>
5  void bubbleSort(T arr[], int n) {
6      for(int i=0;i<n-1;i++)
7          for(int j=0;j<n-i-1;j++)
8              if(arr[j] > arr[j+1])
9                  swap(arr[j], arr[j+1]);
10 }
11
12 int main() {
13     int arr[] = {5,3,1,4};
14     bubbleSort(arr,4);
15
16     for(int i=0;i<4;i++)
17         cout << arr[i] << " ";
18 }
```

```
input
1 3 4 5
...Program finished with exit code 0
Press ENTER to exit console.
```

4. Write a function template to perform linear search.

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  template <typename T>
5  int linearSearch(T arr[], int n, T key) {
6      for(int i=0;i<n;i++)
7          if(arr[i] == key)
8              return i;
9      return -1;
10 }
11
12 int main() {
13     int arr[] = {10,20,30,40};
14     cout << linearSearch(arr,4,30);
15 }
```

input

```
2
...Program finished with exit code 0
Press ENTER to exit console.
```

Class Template

1. Develop a class template for stack with push and pop operations.

main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  template <typename T>
5  class Stack {
6      T arr[100];
7      int top;
8  public:
9      Stack() { top = -1; }
10
11     void push(T x) {
12         arr[++top] = x;
13     }
14
15     void pop() {
16         if(top >= 0) top--;
17     }
18
19     void display() {
20         for(int i=top;i>=0;i--)
21             cout << arr[i] << " ";
22     }
23 };
24
25 int main() {
26     Stack<int> s;
27     s.push(10);
28     s.push(20);
29     s.display();
30 }
```

input

20 10

...Program finished with exit code 0
Press ENTER to exit console.

2. Develop a class template for queue with enqueue and dequeue operations.

```
main.cpp
2  using namespace std;
3
4  template <typename T>
5  class Queue {
6      T arr[100];
7      int front, rear;
8  public:
9      Queue() { front = rear = -1; }
10
11     void enqueue(T x) {
12         if(front == -1) front = 0;
13         arr[++rear] = x;
14     }
15
16     void dequeue() {
17         if(front <= rear) front++;
18     }
19
20     void display() {
21         for(int i=front; i<=rear; i++)
22             cout << arr[i] << " ";
23     }
24 };
25
26 int main() {
27     Queue<int> q;
28     q.enqueue(10);
29     q.enqueue(20);
30     q.display();
31 }
```

input

```
10 20
...Program finished with exit code 0
Press ENTER to exit console.
```

3. Create a class template to store and display a pair of values.

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  template <typename T, typename U>
5  class Pair {
6      T a;
7      U b;
8  public:
9      Pair(T x, U y) {
10         a = x;
11         b = y;
12     }
13
14     void display() {
15         cout << a << " " << b;
16     }
17 };
18
19 int main() {
20     Pair<int, string> p(10, "Hello");
21     p.display();
22 }
```

input

```
10 Hello
...Program finished with exit code 0
Press ENTER to exit console.
```

4. Create a class template for basic arithmetic operations.

ain.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  template <typename T>
5  class Arithmetic {
6      T a, b;
7  public:
8      Arithmetic(T x, T y) {
9          a = x;
10         b = y;
11     }
12
13     void operations() {
14         cout << "Add: " << a+b << endl;
15         cout << "Sub: " << a-b << endl;
16         cout << "Mul: " << a*b << endl;
17         cout << "Div: " << a/b << endl;
18     }
19 };
20
21 int main() {
22     Arithmetic<int> obj(10,5);
23     obj.operations();
24 }
```

input

```
Add: 15
Sub: 5
Mul: 50
Div: 2

...Program finished with exit code 0
Press ENTER to exit console.
```

